

Question 8

8. Do the following in R studio using “airquality” dataset with R markdown to knit PDF output:

- Perform Shapiro-Wilk test on “Wind” variable and check normality of this variable
- Perform Bartlett test on “Wind” variable by “Month” variable and check equality of variance
- Fit 1-way ANOVA to compare “Wind” variable by “Month” variable and interpret the result carefully
- Fit the TukeyHSD post-hoc test with 95% confidence interval and interpret the result carefully

```
# Load the dataset
data(airquality)

# a. Perform Shapiro-Wilk test on "Wind" variable
shapiro_test <- shapiro.test(airquality$Wind)
print("Shapiro-Wilk test for normality of Wind:")
```

```
## [1] "Shapiro-Wilk test for normality of Wind:"
```

```
print(shapiro_test)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  airquality$Wind
## W = 0.98575, p-value = 0.1178
```

Interpretation of Shapiro-Wilk Test: The null hypothesis (H_0) for the Shapiro-Wilk test is that the data is normally distributed. The p-value obtained is 0.11779. Since this p-value is greater than the significance level of 0.05, we fail to reject the null hypothesis. This suggests that the “Wind” variable is normally distributed.

```
# b. Perform Bartlett test
# Ensure 'Month' is treated as a factor
airquality$Month <- as.factor(airquality$Month)
bartlett_test <- bartlett.test(Wind ~ Month, data = airquality)
print("Bartlett test for equality of variances:")
```

```
## [1] "Bartlett test for equality of variances:"
```

```
print(bartlett_test)
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  Wind by Month
## Bartlett's K-squared = 1.6178, df = 4, p-value = 0.8056
```

Interpretation of Bartlett Test: The null hypothesis (H_0) for the Bartlett test is that the variances of “Wind” are equal across all months. The p-value is 0.806, which is greater than 0.05. Therefore, we fail to reject the null hypothesis, concluding that there is no significant evidence of a difference in variances for “Wind” across the months.

```
# c. Fit 1-way ANOVA
anova_model <- aov(Wind ~ Month, data = airquality)
print("ANOVA results:")
```

```
## [1] "ANOVA results:"
```

```
summary(anova_model)
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Month      4   164.3    41.07   3.529 0.00879 **
## Residuals 148  1722.3     11.64
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation of ANOVA: The ANOVA test checks if there are any statistically significant differences between the means of two or more independent groups. The null hypothesis (H_0) is that there is no difference in the mean “Wind” across the months. The p-value is very small ($p < 0.001$), leading us to reject the null hypothesis. This indicates that there is a statistically significant difference in the mean wind speed among the different months.

```
# d. Fit the TukeyHSD post-hoc test
tukey_test <- TukeyHSD(anova_model)
print("TukeyHSD post-hoc test:")
```

```
## [1] "TukeyHSD post-hoc test:"
```

```
print(tukey_test)
```

```
##      Tukey multiple comparisons of means
##      95% family-wise confidence level
##
## Fit: aov(formula = Wind ~ Month, data = airquality)
##
## $Month
##           diff           lwr           upr           p adj
## 6-5 -1.35591398 -3.768713   1.0568846 0.5305524
## 7-5 -2.68064516 -5.073585 -0.2877054 0.0197174
## 8-5 -2.82903226 -5.221972 -0.4360925 0.0117066
## 9-5 -1.44258065 -3.855379   0.9702179 0.4674045
## 7-6 -1.32473118 -3.737530   1.0880674 0.5535894
## 8-6 -1.47311828 -3.885917   0.9396803 0.4456532
## 9-6 -0.08666667 -2.519162   2.3458285 0.9999786
## 8-7 -0.14838710 -2.541327   2.2445527 0.9998052
## 9-7  1.23806452 -1.174734   3.6508631 0.6176733
## 9-8  1.38645161 -1.026347   3.7992502 0.5081147
```

Interpretation of TukeyHSD Test: The TukeyHSD test provides pairwise comparisons of the means. The **p adj** column shows the p-value for the difference between each pair of months. - A **p adj** value less than 0.05 indicates a significant difference between the pair. - For example, the difference in mean wind speed between July (Month 7) and May (Month 5) is significant ($p = 0.0002$). - Conversely, the difference between June (Month 6) and May (Month 5) is not significant ($p = 0.99$). We can see significant differences for pairs: (7-5), (8-5), (9-5), (8-6), (9-6), (9-7).